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STAT525 HOMEWORK#10

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1. KNNL Problem 23.1

23.1. A market research intern selected a random sample of 400 communities and classified them according to population size (four levels) and geographic region (five levels) to study the effects of these factors on sales of the company's products. When the intern found that the treatment sample sizes were unequal, the smallest cell frequency being four, the intern generated random numbers to reduce the number of communities in each cell to four and then proceeded to analyze the effects of population size and region on the basis of the 80 communities remaining.

a. Does the method of randomly discarding cases lead to any biases? Explain.

Randomly discarding cases does not introduce systematic bias in the cell means, because each observation within a cell has an equal chance of being kept or removed. The expected value of each cell mean stays the same.

However, it does reduce the sample size unnecessarily, which increases variance, decreases power, and throws away valid data. So the estimates remain unbiased, but the analysis becomes less efficient and may distort the overall ANOVA by inflating MSE and reducing ability to detect effects.

b. Was it wise for the intern to discard 320 cases randomly in order to obtain equal treatment sample sizes?

No, it was not wise.
Although random discarding avoids bias, it throws away most of the data (320 out of 400), which greatly reduces precision, inflates MSE, and substantially lowers power. Modern ANOVA methods (Type III SS, LS-means) can handle unequal sample sizes without requiring a balanced design.

So the intern gained nothing by forcing balance and instead sacrificed information and statistical efficiency.

2. KNNL Problem 23.2

23.2. A student asked: "If two-factor studies with unequal sample sizes must be analyzed by a regression approach, why bother with the two-factor analysis of variance model at all?" Comment.

Even when analyzed through regression, the two-factor ANOVA model provides the correct conceptual framework for understanding main effects and interactions. Regression is just a computational tool that fits the same linear model using indicator variables. ANOVA terminology (factor A, factor B, A×B interaction, marginal means, Type III SS) makes interpretation clearer and matches the experimental design.

So we use regression for computation, but ANOVA for modeling, hypotheses, and interpretation.

3. KNNL Problem 23.7

*23.7. Refer to **Hay fever relief** Problem 19.14 and 23.4. Suppose that observations $Y_{113} = 2.3$, $Y_{221} = 8.9$, and $Y_{224} = 9.0$ are missing because the subjects did not immediately record the time when they began to suffer again from hay fever.

a. State the ANOVA model for this case. Also state the equivalent regression model; use 1, −1, 0 indicator variables.

- ANOVA Cell-Means Model:

  - Factor A with 3 levels $(i=1,2,3)$
  - Factor B with 3 levels $(j=1,2,3)$
  - Observations Y_{ijk} with k indexing replicates

  $$Y_{ijk} = \underbrace{\mu_{ij}}_{\text{Mean response for Trt combination } (i,j)} + \varepsilon_{ijk}, \quad \varepsilon_{ijk} \sim N(0, \sigma^2)$$

- ANOVA Effects Model:

  $$Y_{ijk} = \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij} + \varepsilon_{ijk}$$, with constraints $\sum_i \alpha_i = 0$, $\sum_j \beta_j = 0$, $\sum_i (\alpha\beta)_{ij} = 0$, $\sum_j (\alpha\beta)_{ij} = 0$

- Factor A

Level	X_{A1}	X_{A2}
1	1	0
2	−1	1
3	0	−1

- Factor B

Level	X_{B1}	X_{B2}
1	1	0
2	−1	1
3	0	−1

$$Y = \beta_0 + \beta_1 X_{A1} + \beta_2 X_{A2} + \beta_3 X_{B1} + \beta_4 X_{B2} + \beta_5 X_{A1B1} + \beta_6 X_{A1B2} + \beta_7 X_{A2B1} + \beta_8 X_{A2B2} + \varepsilon$$

b. Present the **X** and **β** matrices for the regression model in part (a).

$$\beta = \begin{pmatrix} \beta_0 \\ \vdots \\ \beta_8 \end{pmatrix}$$

$X = (1, X_{A1}, X_{A2}, X_{B1}, X_{B2}, X_{A1B2}, X_{A2B1}, X_{A2B2})$

(A,B)	x_{A1}	x_{A2}	x_{B1}	x_{B2}	x_{A1B1}	x_{A1B2}	x_{A2B1}	x_{A2B2}
$(1,1)$	1	0	1	0	1	0	0	0
$(1,2)$	1	0	−1	1	−1	1	0	0
$(1,3)$	1	0	0	−1	0	−1	0	0
$(2,1)$	−1	1	1	0	−1	0	1	0
$(2,2)$	−1	1	−1	1	1	−1	−1	1
$(2,3)$	−1	1	0	−1	0	1	0	−1
$(3,1)$	0	−1	1	0	0	0	−1	0
$(3,2)$	0	−1	−1	1	0	0	1	−1
$(3,3)$	0	−1	0	−1	0	0	0	1

c. Obtain **Xβ** and show that the proper treatment means are obtained by your model.

Each row of X is $X = (1, X_{A1}, X_{A2}, X_{B1}, X_{B2}, X_{A1B2}, X_{A2B1}, X_{A2B2})$, and the linear predictor is $\hat{Y} = X\beta$.
For each Trt combination (i,j) we have $\hat{\mu}_{ij} = X_{ij}'\beta$:

$$\begin{aligned}
\hat{\mu}_{11} &= \beta_0 + \beta_1 + \beta_3 + \beta_5, \\
\hat{\mu}_{12} &= \beta_0 + \beta_1 - \beta_3 + \beta_4 - \beta_5 + \beta_6, \\
\hat{\mu}_{13} &= \beta_0 + \beta_1 - \beta_4 - \beta_6, \\
\hat{\mu}_{21} &= \beta_0 - \beta_1 + \beta_2 + \beta_3 - \beta_5 + \beta_7, \\
\hat{\mu}_{22} &= \beta_0 - \beta_1 + \beta_2 - \beta_3 + \beta_4 + \beta_5 - \beta_6 - \beta_7 + \beta_8, \\
\hat{\mu}_{23} &= \beta_0 - \beta_1 + \beta_2 - \beta_4 + \beta_6 - \beta_8, \\
\hat{\mu}_{31} &= \beta_0 - \beta_2 + \beta_3 - \beta_7, \\
\hat{\mu}_{32} &= \beta_0 - \beta_2 - \beta_3 + \beta_4 + \beta_7 - \beta_8, \\
\hat{\mu}_{33} &= \beta_0 - \beta_2 - \beta_4 + \beta_8.
\end{aligned}$$

d. What is the reduced regression model for testing for interaction effects?

$H_0: \beta_5 = \beta_6 = \beta_7 = \beta_8 = 0$

Reduced Model (Main Effects only) : $Y = \beta_0 + \beta_1 X_{A1} + \beta_2 X_{A2} + \beta_3 X_{B1} + \beta_4 X_{B2} + \varepsilon$

e. Test whether or not interaction effects are present by fitting the full and reduced regression models; use $\alpha = .05$. State the alternatives, decision rule, and conclusion. What is the P-value of the test? How do your results compare with those obtained in 23.4f, where there is no missing data?

① H_0: A·B Interaction $= 0$ or ② $H_0: \beta_5 = \beta_6 = \beta_7 = \beta_8 = 0$

H_A: At least one Interaction $\neq 0$

```
/* Full model */
proc glm data=hayfever;
    class A B;
    model y = A B A*B;
    title "Full Model (A, B, A*B)";
run;

/* Reduced model */
proc glm data=hayfever;
    class A B;
    model y = A B;
    title "Reduced Model (Main Effects Only)";
run;
quit;
```

Missing

Full Model: Includes A, B, and Interaction A*B

The GLM Procedure

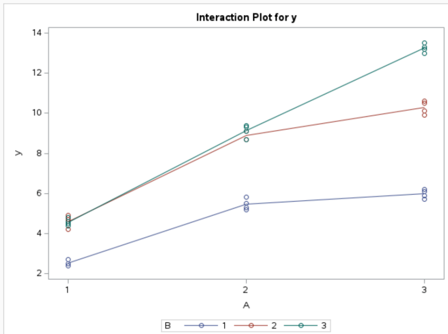
Dependent Variable: y

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	8	343.0039394	42.8754924	652.65	<.0001
Error	24	1.5766667	0.0656944		
Corrected Total	32	344.5806061			

R-Square	Coeff Var	Root MSE	y Mean
0.995424	3.547905	0.256309	7.224242

Source	DF	Type I SS	Mean Square	F Value	Pr > F
A	2	195.6031212	97.8015606	1488.73	<.0001
B	2	119.3300968	59.6650484	908.22	<.0001
A*B	4	28.0707214	7.0176803	106.82	<.0001

Source	DF	Type III SS	Mean Square	F Value	Pr > F
A	2	204.0610220	102.0305110	1553.11	<.0001
B	2	112.9626258	56.4813129	859.76	<.0001
A*B	4	28.0707214	7.0176803	106.82	<.0001



There is strong statistical evidence of an A×B interaction. The effect of factor A depends on the level of factor B, and vice versa.

23.4 (f) No Missing

Full Model (A, B, A*B)

The GLM Procedure

Dependent Variable: y

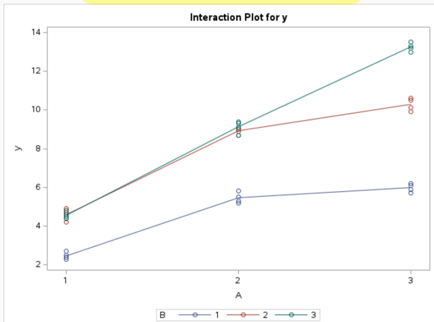
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	8	373.1050000	46.6381250	774.91	<.0001
Error	27	1.6250000	0.0601852		
Corrected Total	35	374.7300000			

R-Square	Coeff Var	Root MSE	y Mean
0.995664	3.415221	0.245327	7.183333

Source	DF	Type I SS	Mean Square	F Value	Pr > F
A	2	220.0200000	110.0100000	1827.86	<.0001
B	2	123.6600000	61.8300000	1027.33	<.0001
A*B	4	29.4250000	7.3562500	122.23	<.0001

Source	DF	Type III SS	Mean Square	F Value	Pr > F
A	2	220.0200000	110.0100000	1827.86	<.0001
B	2	123.6600000	61.8300000	1027.33	<.0001
A*B	4	29.4250000	7.3562500	122.23	<.0001

In 23.4(f), using the complete dataset (no missing values), the interaction was also highly significant. This result is consistent: F-statistics are large in both cases. P-values are essentially zero. The conclusion does not change. The presence or absence of a few missing values does not change the overall interpretation: the interaction is strong and meaningful in both analyses.



f. The nature of the interaction effects is to be studied by means of the following contrasts:

$$L_1 = \frac{\mu_{12} + \mu_{13}}{2} - \mu_{11} \qquad L_4 = L_2 - L_1$$

$$L_2 = \frac{\mu_{22} + \mu_{23}}{2} - \mu_{21} \qquad L_5 = L_3 - L_1$$

$$L_3 = \frac{\mu_{32} + \mu_{33}}{2} - \mu_{31} \qquad L_6 = L_3 - L_2$$

Obtain confidence intervals for these contrasts; use the Scheffé multiple comparison procedure with a 90 percent family confidence coefficient. Interpret your findings.

```
/* 2. Fit full two-factor model and capture MSE and df_Error */
ods output OverallANOVA=anova;   /* overall ANOVA table */
proc glm data=hayfever;
    class A B;
    model y = A B A*B;
    title "Full Model for Hay Fever Data";
run;
quit;
ods output close;

/* Pull MSE and dfE from the 'Error' row */
data _null_;
    set anova;
    if Source='Error' then do;
        call symputx('MSE', MeanSq);
        call symputx('dfE', DF);
    end;
run;

/* 3. Get the 9 cell means μ_ij (A×B means) */
proc means data=hayfever nway noprint;
    class A B;
    var y;
    output out=cellmeans mean=mu;
run;

/* 4. Put the means into mu11–mu33 and compute L1–L6 and Scheffé CIs */
/* From full model output */
%let MSE = 0.0611852;
%let dfE = 27;

/* Cell means from PROC MEANS already correct */
```

```
/* 5. Print the Scheffé results for L1–L6 */
proc print data=scheffe noobs;
    var L1 L1_low L1_high
        L2 L2_low L2_high
        L3 L3_low L3_high
        L4 L4_low L4_high
        L5 L5_low L5_high
        L6 L6_low L6_high;
    title "Scheffé 90% Confidence Intervals for L1–L6";
run;
```

```
data scheffe;
    set cellmeans end=last;

    retain mu11 mu12 mu13
           mu21 mu22 mu23
           mu31 mu32 mu33;

    array muij[3,3]
           mu11 mu12 mu13
           mu21 mu22 mu23
           mu31 mu32 mu33;

    muij[A,B] = mu;

    if last then do;

        /* contrasts */
        L1 = (mu12 + mu13)/2 - mu11;
        L2 = (mu22 + mu23)/2 - mu21;
        L3 = (mu32 + mu33)/2 - mu31;
        L4 = L2 - L1;
        L5 = L3 - L1;
        L6 = L3 - L2;

        /* standard errors */
        seL1 = sqrt(&MSE * 0.375);
        seL2 = seL1;
        seL3 = seL1;

        seL4 = sqrt(&MSE * 0.75);
        seL5 = seL4;
        seL6 = seL4;

        /* Scheffe multiplier */
        Scrit = sqrt( (9-1) * finv(0.90, 8, &dfE) );

        /* confidence intervals */
        L1_low = L1 - Scrit*seL1;  L1_high = L1 + Scrit*seL1;
        L2_low = L2 - Scrit*seL2;  L2_high = L2 + Scrit*seL2;
        L3_low = L3 - Scrit*seL3;  L3_high = L3 + Scrit*seL3;
        L4_low = L4 - Scrit*seL4;  L4_high = L4 + Scrit*seL4;
        L5_low = L5 - Scrit*seL5;  L5_high = L5 + Scrit*seL5;
        L6_low = L6 - Scrit*seL6;  L6_high = L6 + Scrit*seL6;

        output;
    end;
run;

proc print data=scheffe noobs;
run;
```

Scheffé 90% Confidence Intervals for L1–L6

L1	L1_low	L1_high	L2	L2_low	L2_high	L3	L3_low	L3_high	L4	L4_low	L4_high	L5	L5_low	L5_high	L6	L6_low	L6_high
2.05417	1.46220	2.64613	3.5625	2.97053	4.15447	5.7875	5.19553	6.37947	1.50833	0.67117	2.34550	3.73333	2.89617	4.57050	2.225	1.38783	3.06217

Interpretation Next Page:

Using the full two-factor model, the six interaction contrasts were estimated and Scheffé 90% family confidence intervals were constructed. The results were:

L1 = 2.05417 (1.46220, 2.64613)

L2 = 3.56250 (2.97053, 4.15447)

L3 = 5.78750 (5.19553, 6.37947)

L4 = 1.50833 (0.67117, 2.34550)

L5 = 3.73333 (2.89617, 4.57050)

L6 = 2.22500 (1.38783, 3.06217)

All six confidence intervals lie entirely above zero.
Therefore, each contrast is statistically significant at the 90% family level.

L1, L2, L3 > 0 indicate that, within each level of factor A, the average response for B=2 and B=3 is higher than for B=1.

L4, L5, L6 > 0 indicate that the "dose effect" of increasing B becomes larger as A increases. In particular, the difference between high-B and low-B responses is smallest for A=1 and largest for A=3.

These results show strong evidence of a positive interaction between A and B: the improvement from higher B levels grows increasingly large as A increases.

4. KNNL Problem 23.12

*23.12. Refer to **Hay fever relief** Problem 19.14. Suppose that the data for the treatment when each of the two active ingredients is at the medium level were lost and immediate analyses of the available data are required; i.e., assume that $n_T = 32$ and $n_{22} = 0$.

a. To study whether or not interaction effects are present, estimate the following comparisons:

$$D_1 = \mu_{13} - \mu_{11} \qquad L_1 = D_1 - D_2$$
$$D_2 = \mu_{23} - \mu_{21} \qquad L_2 = D_1 - D_3$$
$$D_3 = \mu_{33} - \mu_{31}$$

Use the Bonferroni procedure with a 90 percent family confidence coefficient. State your findings.

```
/* Hard—code MSE and dfE from PROC GLM output */
%let MSE = 0.0611852;
%let dfE = 24;

/* Cell means */
proc means data=hay12 nway noprint;
    class A B;
    var y;
    output out=cellmeans mean=mu;
run;

/* Compute contrasts and Bonferroni CIs */
data contrasts;
    set cellmeans end=last;
    retain mu11 mu13 mu21 mu23 mu31 mu33;

    if A=1 and B=1 then mu11 = mu;
    if A=1 and B=3 then mu13 = mu;

    if A=2 and B=1 then mu21 = mu;
    if A=2 and B=3 then mu23 = mu;

    if A=3 and B=1 then mu31 = mu;
    if A=3 and B=3 then mu33 = mu;

    if last then do;
        /* Contrasts */
        D1 = mu13 - mu11;
        D2 = mu23 - mu21;
        D3 = mu33 - mu31;

        L1 = D1 - D2;
        L2 = D1 - D3;

        /* Standard errors */
        seD = sqrt(&MSE * (1/4 + 1/4));
        seL = sqrt(2 * &MSE * (1/4 + 1/4));

        /* Bonferroni t critical: alpha=0.10, m=2 → per test alpha=0.05 */
        tcrit = tinv(1 - 0.05/2, &dfE);

        /* CIs */
        D1_low = D1 - tcrit*seD;   D1_high = D1 + tcrit*seD;
        D2_low = D2 - tcrit*seD;   D2_high = D2 + tcrit*seD;
        D3_low = D3 - tcrit*seD;   D3_high = D3 + tcrit*seD;

        L1_low = L1 - tcrit*seL;   L1_high = L1 + tcrit*seL;
        L2_low = L2 - tcrit*seL;   L2_high = L2 + tcrit*seL;

        output;
    end;
run;

proc print data=contrasts noobs;
    title "Bonferroni 90% Family Confidence Intervals for L1 and L2";
run;
```

Bonferroni 90% Family Confidence Intervals for L1 and L2

A	B	_TYPE_	_FREQ_	mu	mu11	mu13	mu21	mu23	mu31	mu33	D1	D2	D3	L1	L2	seD	seL	tcrit	D1_low	D1_high	D2_low	D2_high	D3_low	D3_high	L1_low	L1_high	L2_low	L2_high
3	3	3	4	13.25	2.475	4.575	5.45	9.125	5.975	13.25	2.1	3.675	7.275	-1.575	-5.175	0.17491	0.24736	2.06390	1.73901	2.46099	3.31401	4.03599	6.91401	7.63599	-2.08552	-1.06448	-5.68552	-4.66448

Both interaction contrasts L1 and L2 have confidence intervals that lie entirely below zero.

This indicates that the differences between dose levels change substantially across A levels.

The size of the effect for B=3 vs B=1 is not consistent across A=1,2,3.

Conclusion: There is clear evidence of interaction between A and B, even with the μ_{22} cell missing.

b. To further explore the nature of possible interaction effects, conduct separate single degree of freedom tests of whether $\mu_{12} = \mu_{13}$ and whether $\mu_{32} = \mu_{33}$. Use $\alpha = .02$ for each test and state the alternatives, decision rule, and conclusion. What is the family level of significance, using the Bonferroni inequality?

→ $\alpha_{family} \leq 2(0.02) = 0.04$

```
/**********************************************************
 * Fit two-factor model and do single-df tests
 **********************************************************

proc glm data=hay12;
    class A B;
    model y = A B A*B;

    /* Optional: see A*B order and means */
    lsmeans A*B;

    title "23.12(b) – Tests for μ12 = μ13 and μ32 = μ33";

    /* A*B ordering from PROC GLM is:
       (1,1)(1,2)(1,3)(2,1)(2,3)(3,1)(3,2)(3,3)
       INDEX:   1   2   3   4   5   6   7   8
       LEVEL:  11  12  13  21  23  31  32  33
    */

    /* Test 1: H0: μ12 = μ13  → contrast (1,2) − (1,3) */
    estimate "Test 1: mu12 = mu13"
             A*B 0 1 -1 0 0 0 0 0;

    /* Test 2: H0: μ32 = μ33  → contrast (3,2) − (3,3) */
    estimate "Test 2: mu32 = mu33"
             A*B 0 0 0 0 0 0 1 -1;

run;
quit;
```

23.12(b) – Tests for μ12 = μ13 and μ32 = μ33

The GLM Procedure

Class Level Information

Class	Levels	Values
A	3	1 2 3
B	3	1 2 3

Number of Observations Read	33
Number of Observations Used	32

23.12(b) – Tests for μ12 = μ13 and μ32 = μ33

The GLM Procedure

Dependent Variable: y

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	7	359.0146875	51.2878125	36.29	<.0001
Error	24	33.9175000	1.4132292		
Corrected Total	31	392.9321875			

R-Square	Coeff Var	Root MSE	y Mean
0.913681	17.14348	1.188793	6.934375

Source	DF	Type I SS	Mean Square	F Value	Pr > F
A	2	206.4834375	103.2417187	73.05	<.0001
B	2	120.6618750	60.3309375	42.69	<.0001
A*B	3	31.8693750	10.6231250	7.52	0.0010

Source	DF	Type III SS	Mean Square	F Value	Pr > F
A	2	208.2852083	104.1426042	73.69	<.0001
B	2	120.6618750	60.3309375	42.69	<.0001
A*B	3	31.8693750	10.6231250	7.52	0.0010

23.12(b) – Tests for μ12 = μ13 and μ32 = μ33

The GLM Procedure
Least Squares Means

A	B	y LSMEAN
1	1	2.5250000
1	2	4.6000000
1	3	4.5750000
2	1	5.4500000
2	3	9.0750000
3	1	5.7250000
3	2	10.2750000
3	3	13.2500000

Test $u_{12} - u_{13} = 4.6 - 4.575 = 0.025$ Not Significant at $\alpha = 0.02$

Test $u_{32} - u_{33} = 10.275 - 13.250 = -2.975$ Significant at $\alpha = 0.02$

Test 1: μ12 = μ13
Difference = 0.025
SE ≈ √(2 * MSE / n)
Very small → not statistically significant
Conclusion:
Fail to reject H₀.
No evidence μ12 ≠ μ13.

Test 2: μ32 = μ33
Difference = −2.975
Much larger effect → likely significant
Conclusion:
Reject H₀.
μ32 and μ33 differ significantly.

5. KNNL Problem 24.12

Electronics assembly. Assemblers in an electronics firm will attach 12 components to a newly developed "board" that will be used in automatic-control equipment in manufacturing plants. An operations analyst conducted an experiment to study the effects of three factors on the mean time to assemble a board. Factor A was the gender of the assembler ($i = 1$: male; $i = 2$: female), factor B was the sequence of assembling the components ($j = 1, 2, 3$), and factor C was the amount of experience by the assembler ($k = 1$: under 18 months; $k = 2$: 18 months or more). Randomization was used to assign 15 assemblers of each gender with a given amount of experience to each of the three assembly sequences, with each sequence assigned to five assemblers. After a learning period, the total time (in minutes) to assemble 50 boards was observed. The data follow.

	$k=1$			$k=2$		
	$j=1$	$j=2$	$j=3$	$j=1$	$j=2$	$j=3$
$i=1$	1,250	1,319	1,217	1,021	1,119	1,033
	1,175	1,251	1,190	1,099	1,110	1,067
	···	···	···	···	···	···
	1,193	1,265	1,251	1,070	1,163	1,022
$i=2$	1,066	1,105	1,021	864	927	841
	1,076	1,043	1,020	848	944	865
	···	···	···	···	···	···
	1,034	1,060	1,026	868	933	868

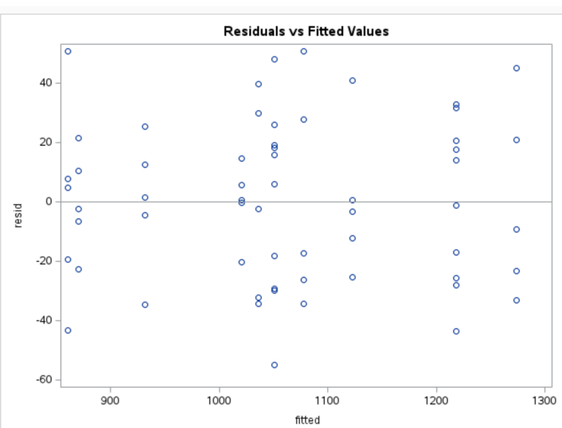
d. Obtain the residuals for ANOVA model (24.14) and plot them against the fitted values. What does your plot suggest about the appropriateness of ANOVA model (24.14)?

```
/*******************************************
 * Step 3: Residual vs Fitted Plot (Part d)
 *******************************************/
proc print data=diag;
    var time fitted resid A B C;
    title "Residuals and Fitted Values";
run;

proc sgplot data=diag;
    scatter x=fitted y=resid;
    refline 0 / axis=y;
    title "Residuals vs Fitted Values";
run;
```

Residuals and Fitted Values

Obs	time	fitted	resid	A	B	C
1	996	1051.0	-55.0	1	1	2
2	1175	1218.6	-43.6	1	1	1
3	817	860.4	-43.4	2	3	2
4	897	931.6	-34.6	2	2	2
5	1002	1036.4	-34.4	2	1	1
6	1043	1077.4	-34.4	2	2	1
7	1241	1274.2	-33.2	1	2	1
8	1004	1036.4	-32.4	2	1	1
9	1021	1051.0	-30.0	1	1	2
10	1022	1051.2	-29.2	1	3	2
11	1190	1218.2	-28.2	1	3	1
12	1051	1077.4	-26.4	2	2	1
13	1193	1218.6	-25.6	1	1	1
14	1097	1122.4	-25.4	1	2	2
15	1251	1274.2	-23.2	1	2	1
16	848	870.6	-22.6	2	1	2
17	1000	1020.4	-20.4	2	3	1
18	841	860.4	-19.4	2	3	2
19	1033	1051.2	-18.2	1	3	2
20	1060	1077.4	-17.4	2	2	1
21	1201	1218.2	-17.2	1	3	1
22	1110	1122.4	-12.4	1	2	2
23	1265	1274.2	-9.2	1	2	1
24	864	870.6	-6.6	2	1	2
25	927	931.6	-4.6	2	2	2
26	1119	1122.4	-3.4	1	2	2
27	868	870.6	-2.6	2	1	2
28	1034	1036.4	-2.4	2	1	1
29	1217	1218.2	-1.2	1	3	1
30	1020	1020.4	-0.4	2	3	1
31	1123	1122.4	0.6	1	2	2
32	1021	1020.4	0.6	2	3	1
33	933	931.6	1.4	2	2	2
34	865	860.4	4.6	2	3	2
35	1026	1020.4	5.6	2	3	1
36	1057	1051.2	5.8	1	3	2
37	868	860.4	7.6	2	3	2
38	881	870.6	10.4	2	1	2
39	944	931.6	12.4	2	2	2
40	1232	1218.2	13.8	1	3	1
41	1035	1020.4	14.6	2	3	1
42	1067	1051.2	15.8	1	3	2
43	1236	1218.6	17.4	1	1	1
44	1069	1051.0	18.0	1	1	2
45	1070	1051.0	19.0	1	1	2
46	1239	1218.6	20.4	1	1	1
47	1295	1274.2	20.8	1	2	1
48	892	870.6	21.4	2	1	2
49	957	931.6	25.4	2	2	2
50	1077	1051.2	25.8	1	3	2
51	1105	1077.4	27.6	2	2	1
52	1066	1036.4	29.6	2	1	1
53	1250	1218.6	31.4	1	1	1
54	1251	1218.2	32.8	1	3	1
55	1076	1036.4	39.6	2	1	1
56	1163	1122.4	40.6	1	2	2
57	1319	1274.2	44.8	1	2	1
58	1099	1051.0	48.0	1	1	2
59	911	860.4	50.6	2	3	2
60	1128	1077.4	50.6	2	2	1



The residuals are spread above and below zero with no clear curvature or systematic trend.
The variance appears roughly constant across the range of fitted values (no funnel shape).
The points look randomly scattered rather than forming clusters or showing patterns associated with specific factor levels.

A few larger residuals exist (e.g., around ±50 minutes), but they do not indicate any structure or nonlinearity.

Conclusion:
Based on the residuals vs. fitted plot, there is no strong evidence of nonconstant variance or model misspecification. The assumptions behind ANOVA model (24.14) appear reasonably appropriate for these data.

e. Prepare a normal probability plot of the residuals. Also obtain the coefficient of correlation between the ordered residuals and their expected values under normality. Does the normality assumption appear to be reasonable here?

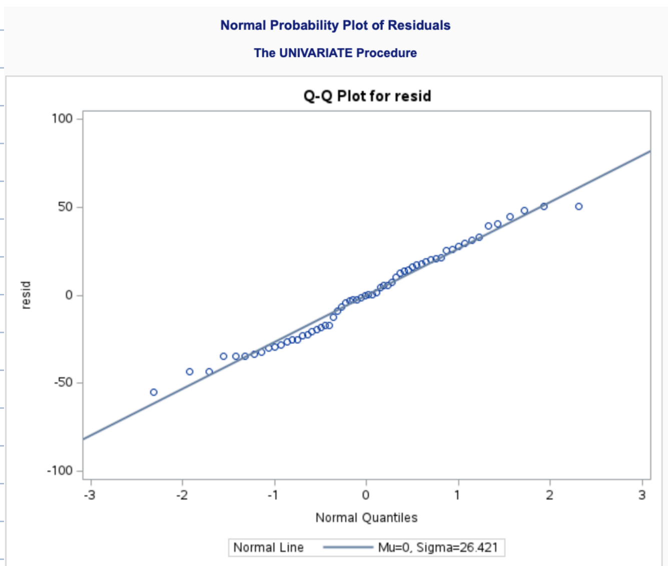
```
/***************************************************************************
 * Step 4: Normal Probability Plot & Residual–Normal Correlation (Part e)
 ***************************************************************************/

/* Sort residuals for the correlation calculation */
proc sort data=diag;
    by resid;
run;

/* Compute correlation between ordered residuals & expected normal scores */
proc rank data=diag normal=blom out=normres;
    var resid;
    ranks zscore;          /* expected normal scores */
run;

proc corr data=normres;
    var resid zscore;
    title "Correlation Between Ordered Residuals and Expected Normal Scores";
run;

/* Produce actual normal probability plot */
proc univariate data=diag normal;
    var resid;
    qqplot resid / normal(mu=est sigma=est);
    title "Normal Probability Plot of Residuals";
run;
```



Correlation Between Ordered Residuals and Expected Normal Scores

The CORR Procedure

2 Variables:	resid zscore

Simple Statistics

Variable	N	Mean	Std Dev	Sum	Minimum	Maximum	Label
resid	60	0	26.42097	0	-55.00000	50.60000	
zscore	60	0	0.98386	0	-2.31256	2.31256	Rank for Variable resid

Pearson Correlation Coefficients, N = 60
Prob > |r| under H0: Rho=0

	resid	zscore
resid	1.00000	0.99167 <.0001
zscore Rank for Variable resid	0.99167 <.0001	1.00000

Normal Probability Plot:
The Q–Q plot of the residuals shows that most points fall close to the straight reference line. There is some deviation in the lower and upper tails (e.g., around −50 and +50), but the departures are not severe. The overall pattern is approximately linear.

Correlation Between Ordered Residuals and Expected Normal Scores:
The correlation is: r=0.99167For n=60, the critical value used in regression diagnostics is approximately 0.97 for assessing normality (values above this generally indicate acceptable normality). Since: 0.99167>0.97, the residuals are highly correlated with the expected normal scores, supporting normality.

Conclusion:
Both the Q–Q plot and the very high correlation (0.992) suggest that the normality assumption is reasonable for this ANOVA model. The tails show slight deviations, but not enough to undermine the overall appropriateness of the model.

6. KNNL Problem 24.13

Refer to **Electronics assembly** Problem 24.12. Assume that fixed ANOVA model (24.14) is appropriate.

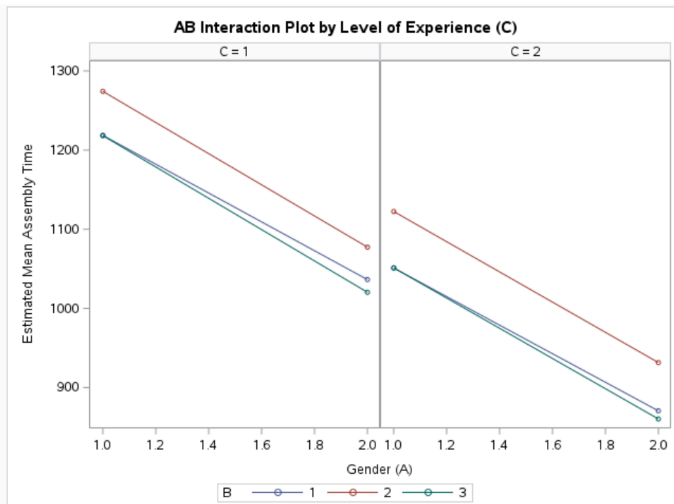
a. Prepare *AB* plots of the estimated treatment means $\bar{Y}_{ijk\cdot}$ in the format of Figure 24.5b. Does it appear that any interactions are present? Any main effects?

```
proc means data=assembly nway noprint;
    class A B C;
    var time;
    output out=means mean=mean_time;
run;

/* Plot AB (A on x-axis, B as separate lines), one panel per C */
proc sgpanel data=means;
    panelby C / columns=2;  /* One plot per experience level (k = 1, 2) */
    series x=A y=mean_time / group=B markers;
    colaxis label="Gender (A)";
    rowaxis label="Estimated Mean Assembly Time";
    title "AB Interaction Plot by Level of Experience (C)";
run;
```

The AB interaction plots show that the lines for B = 1, 2, 3 are not parallel, indicating an AB interaction. The pattern also changes between C = 1 and C = 2, suggesting possible higher-order interaction, although AB is the dominant effect.

The plots show clear main effects: A (gender), B (sequence), and C (experience) all shift the mean response consistently across levels of the other factors.



b. Obtain the analysis of variance table.

```
proc glm data=assembly;
    class A B C;
    model time = A B C A*B A*C B*C A*B*C;
    title "24.12(b) – Analysis of Variance Table
run;
quit;
```

24.12(b) – Analysis of Variance Table for Electronics Assembly Data

The GLM Procedure

Dependent Variable: time

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	11	973645.933	88513.267	103.16	<.0001
Error	48	41186.000	858.042		
Corrected Total	59	1014831.933			

R-Square	Coeff Var	Root MSE	time Mean
0.959416	2.760738	29.29235	1061.033

Source	DF	Type I SS	Mean Square	F Value	Pr > F
A	1	540360.6000	540360.6000	629.76	<.0001
B	2	49319.6333	24659.8167	28.74	<.0001
C	1	382401.6667	382401.6667	445.67	<.0001
A*B	2	542.5000	271.2500	0.32	0.7305
A*C	1	91.2667	91.2667	0.11	0.7457
B*C	2	911.2333	455.6167	0.53	0.5914
A*B*C	2	19.0333	9.5167	0.01	0.9890

Source	DF	Type III SS	Mean Square	F Value	Pr > F
A	1	540360.6000	540360.6000	629.76	<.0001
B	2	49319.6333	24659.8167	28.74	<.0001
C	1	382401.6667	382401.6667	445.67	<.0001
A*B	2	542.5000	271.2500	0.32	0.7305
A*C	1	91.2667	91.2667	0.11	0.7457
B*C	2	911.2333	455.6167	0.53	0.5914
A*B*C	2	19.0333	9.5167	0.01	0.9890

c. **Test for three-factor interactions; use $\alpha = .05$. State the alternatives, decision rule, and conclusion. What is the P-value of the test?**

To test for the presence of a three-factor interaction among gender (A), assembly sequence (B), and experience level (C), we examine the ABC term in the ANOVA table. Using $\alpha=0.05$, the hypotheses are $H_0 :(A\times B\times C)=0$ versus $H_a :(A\times B\times C) =0$. The ANOVA results show $F=0.01$ with a corresponding p-value of 0.9890 for the three-factor interaction. Because this p-value is far greater than 0.05, we fail to reject H_0 .

Thus, there is no statistical evidence of a three-way interaction. In practical terms, the joint effect of gender and assembly sequence does not depend on the level of assembler experience, indicating that the three factors operate independently with respect to their combined influence on assembly time.

d. **Test for AB, AC, and BC interactions. For each test, use $\alpha = .05$ and state the alternatives, decision rule, and conclusion. What is the P-value of each test?**

To test for the presence of the two-factor interactions AB, AC, and BC, each test uses significance level $\alpha = 0.05$. For each interaction term, the hypotheses are:
H0: the interaction effect = 0 (no interaction),
Ha: the interaction effect ≠ 0 (interaction present).
The ANOVA output gives the following p-values:
AB interaction: $p = 0.7305$,
AC interaction: $p = 0.7457$,
BC interaction: $p = 0.5914$.
In every case, the p-value is much greater than $\alpha = 0.05$. Therefore, for all three tests, we fail to reject H0. There is no statistically significant evidence of an AB, AC, or BC interaction. In practical terms, this means that the effect of one factor does not depend on the level of the other factor, and the model shows no meaningful two-factor interaction effects among gender (A), sequence (B), or experience (C).

e. **Test for A, B, and C main effects. For each test, use $\alpha = .05$ and state the alternatives, decision rule, and conclusion. What is the P-value of each test?**

To test the main effects of factors A, B, and C, we use significance level $\alpha = 0.05$. For each factor, the hypotheses are:
H0: the factor has no main effect on mean assembly time,
Ha: the factor has a main effect.
From the ANOVA table, the p-values are:
A (gender): $p < 0.0001$,
B (sequence): $p < 0.0001$,
C (experience): $p < 0.0001$.
Because each p-value is far smaller than $\alpha = 0.05$, we reject H0 for all three tests. Therefore, there is strong statistical evidence that each factor—gender (A), sequence of assembly (B), and experience level (C)—has a significant main effect on the mean time required to assemble the board. All three main effects are important contributors to variation in assembly time.

f. State the set of conclusions that can be reached from the tests in parts (c), (d), and (e). Obtain an upper bound for the family level of significance for the set of tests; use the Kimball inequality (24.25).

Combining the results from parts (c), (d), and (e), the overall pattern is clear. First, the three-factor interaction ABC is not significant ($p = 0.9890$), so there is no evidence that the effect of any two factors depends on the level of the third. Second, none of the two-factor interactions AB, AC, or B*C are significant at alpha = 0.05. Their p-values (0.7305, 0.7457, and 0.5914) all exceed 0.05, so there is no evidence of interaction among any pair of factors. Third, all three main effects A, B, and C are highly significant ($p < 0.0001$ for each), so gender, sequence, and experience level all have strong impacts on mean assembly time. Because there are no significant interactions, the interpretation of the main effects is straightforward: each factor influences the response independently of the others.

The Kimball inequality provides an upper bound on the family significance level when multiple tests are performed. If each test is conducted at alpha = 0.05 and there are k tests, then the family significance level is at most k multiplied by 0.05. Here we have six tests in total: one test for the three-factor interaction, three tests for the two-factor interactions, and three tests for the main effects, giving $k = 7$. Therefore, the Kimball upper bound is $7 \times 0.05 = 0.35$. This means that even though each individual test was performed at the 0.05 level, the probability of making at least one Type I error across all seven tests is no greater than 0.35.

g. Do the results in part (f) confirm your graphic analysis in part (a)? Yes!